

Mixture-of-Experts in Large Language Models

From Sparse Scaling to Granularity, Balancing, and Serving

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Abstract

Sparsely-activated Mixture-of-Experts (MoE) has moved from a scaling curiosity to the default architecture of frontier open language models, yet the techniques behind that shift are routinely conflated and their benchmark numbers reported without a common frame. Existing surveys catalogue methods by component—routing, granularity, balancing, serving—but stop short of asking which of them now carries the field’s measurable gains. We organise the literature around three levers any MoE design must set: how tokens are *routed* to experts, how experts are *granularised and shared*, and how the resulting model is *balanced and served*, and we read each method through what it changes on those levers rather than through its headline parameter count. Surveying 106 papers, tech reports, and model cards across 2020–2026, we compare the sixteen most influential MoE systems along six normalised dimensions, separating each model’s *total* parameter count from the *active* budget that actually sets its compute. Our headline finding is that MoE’s measurable gains have shifted away from raw expert count and hand-tuned load-balancing toward fine-grained and shared-expert granularity plus auxiliary-loss-free balancing, and that the binding constraint on frontier MoE is now serving-systems efficiency rather than the modelling recipe. The migration of the literature’s centre of gravity toward inference-systems work, the convergence of frontier models onto a shared granularity-plus-balancing recipe, and the flattening of quality gains from expert count at fixed active budget all support this reading and reframe the field’s open problems around memory, routing robustness, and active-budget-controlled evaluation.

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1 Introduction

In 2021, sparsely-activated Mixture-of-Experts (MoE) was an exotic way to reach a trillion parameters [Fedus et al., 2021]; by 2026 it is the default architecture of frontier open models, from Mixtral through DeepSeek-V3 (Figure 2). The promise is simple to state: grow the parameter count—and therefore the knowledge—of a model while holding the per-token compute roughly fixed, because each token activates only a few of many experts. The purpose of this survey is not to catalogue MoE variants in isolation but to ask a sharper question: now that MoE has won, where do its remaining gains actually come from? We answer it by forcing every method onto a single comparison frame and separating what a model claims through its *total* parameter count from what it spends through its *active* budget.

1.1 Why Now?

Three concurrent shifts make this the right moment to take stock. First, routing largely settled: after a burst of alternatives—expert-choice, hashing, balanced assignment—the field converged on learned top- k token-choice, so the router is no longer where designs differ most. Second, granularity and balancing matured into recipes: fine-grained experts with a few always-on shared experts [Dai et al., 2024], and auxiliary-loss-free balancing [Wang et al., 2024], turned two long-standing pain points into commoditized layers. Third, and decisively, serving became the binding economic problem: an MoE’s total parameters must still reside in memory even though only a fraction

are active, so the literature’s centre of gravity has moved to expert offloading and expert-parallel inference [Eliseev and Mazur, 2023, Xue et al., 2024]. We organise the survey around the three levers these shifts expose—**(L1)** routing, **(L2)** granularity and balancing, and **(L3)** serving systems—and derive the taxonomy of Figure 1 from that decomposition rather than asserting it. The same lens reads the field’s trajectory (Figure 2): the dominant lever migrates from raw sparse scaling toward granularity and balancing, and most recently toward serving.

1.2 Relationship to Existing Surveys

Several surveys cover adjacent ground; our delta is the three-lever frame and the total-versus-active axis. Cai et al. [2025] and Cai et al. [2024] catalogue MoE methods comprehensively by component; we embed that catalogue inside a thesis about *which* component now carries the gains. Vats et al. [2024] trace MoE’s history from basics to breakthroughs; we narrow to autoregressive LLMs and add a normalised total-versus-active comparison. Liu et al. [2024] review inference-optimization techniques for MoE, and Yuan et al. [2024b] frame LLM inference through a roofline model; we connect both to the production recipes of Section 5 and argue that serving, not modelling, is the binding constraint. None separates total from active parameters as the organising measurement.

1.3 Thesis and Contributions

Our thesis is that MoE’s measurable gains have shifted away from raw expert count and hand-tuned load-balancing toward fine-grained and shared-expert granularity plus auxiliary-loss-free balancing, and that the binding constraint on frontier MoE is now serving-systems efficiency rather than the modelling recipe. We steelman the opposite—that raw scale (expert and total-parameter count) still drives quality and serving is amortizable engineering—before rejecting it. This survey contributes:

1. **Three-Lever Frame.** We reduce MoE design to routing, granularity/balancing, and serving, and derive a five-tier taxonomy from them, stating for each method what lever it moves and what it trades away (§2).
2. **Total-vs-Active Analysis.** We separate total from active parameters across the corpus and show that quality per active FLOP, not raw count, is the honest axis (§6).
3. **Normalised Comparison Matrix.** We compare sixteen influential MoE systems along six dimensions in a single cross-cutting matrix (§4).
4. **Convergence Account.** We reverse-engineer how frontier MoE models stack routing, granularity, balancing, precision, and serving, and reframe the open problems around memory, routing robustness, and evaluation (§5).

The remainder proceeds through the three levers in turn and closes with paired open problems, future directions, and a re-framing conclusion.

2 Background: Sparse Activation and Three Levers

A Mixture-of-Experts layer is best understood not as a catalogue of routing tricks but as a setting of three levers—how tokens are routed, how experts are granularised and shared, and how the model is balanced and served—and this decomposition, not the conventional component labels, is what later lets us locate the binding constraint.

MoE generation $\times$ feature dimensions

	Dominant lever	Routing	Granularity	Balancing	Serving technique
T1 Sparse scaling...	• sparse scaling	• token-choice top-1/2	• few large experts	• aux loss	• expert parallel
T2 Stabilized (2021-2022)	• stability	• expert-choice, BASE...	• large experts	• aux + z-loss	• capacity factor
T3 Open frontier...	• granularity	• token-choice top-2	• fine-grained + shared	• aux loss	• all-to-all
T4 Production MoE...	• granularity +...	• token-choice top-k	• fine-grained + shared	• auxiliary-loss-free	• FP8, expert parallel
T5 Serving-bound...	• serving systems	• settled	• fine-grained	• loss-free	• offloading, prediction
<i>Key Insight: The dominant lever migrates from raw sparse scaling (T1) to granularity + loss-free balancing (T4) and then to serving systems (T5); routing is settled by T3.</i>					

Figure 1: Comprehensive taxonomy of Mixture-of-Experts in Large Language Models across 5 feature dimensions and 5 tiers (106 papers surveyed).

A component-based taxonomy—router, experts, balancing loss, kernels—is the conventional organization and is arguably more neutral, because it groups methods by what they look like rather than by a causal story about where the gains are. If the three levers were not separable—if, say, granularity could not be changed without re-opening routing—the decomposition would be a rhetorical device rather than an analytic one.

2.1 What sparse activation buys

A dense feed-forward block applies the same parameters to every token. An MoE block replaces it with N expert sub-networks and a router that sends each token to $k \ll N$ of them, so the *total* parameter count grows with N while the *active* count—and therefore the per-token compute—grows only with k . This decoupling is the whole point: Fedus et al. [2021] reach a trillion total parameters while keeping per-token cost close to a far smaller dense model, and Du et al. [2021] show an MoE matching a dense baseline’s quality at a fraction of the activated FLOPs. The catch is equally structural: all N experts must still be *stored*, so an MoE trades the recurring cost of compute for the standing cost of memory—a trade that, as we argue in §5, sets the modern bottleneck.

2.2 L1, L2, L3: a decomposition of the design

(L1) Routing. A router assigns tokens to experts; the choices are the assignment rule (token-choice, expert-choice, hashing, balanced assignment) and the top- k value. **(L2) Granularity and balancing.** Given a parameter budget, a designer chooses few large experts or many fine-grained ones, whether to reserve always-on shared experts, and how to keep load balanced across experts without a quality penalty. **(L3) Serving.** Because total parameters dominate memory,

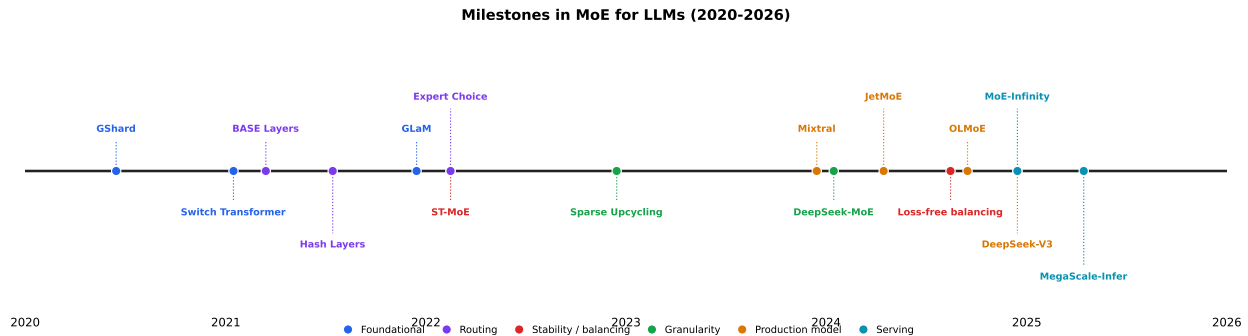


Figure 2: Milestones in Mixture-of-Experts for LLMs (2020–2026), coloured by contribution type. The dominant lever migrates from sparse scaling to granularity and loss-free balancing, and most recently to serving systems.

the deployable model depends on expert parallelism, all-to-all communication, capacity factors, and offloading. The three levers differ in character: L1 is a largely *settled* design choice, L2 is where recent *modelling* gains concentrate, and L3 is a *recurring* deployment cost paid on every forward pass. A field optimizing total cost of ownership will keep migrating effort toward the recurring constraint long after the modelling levers settle—exactly the migration this survey documents.

2.3 Why load balancing is the classic failure

Left unconstrained, a learned router collapses: a few experts attract most tokens, the rest atrophy, and effective capacity falls far below the nominal N . The historical fix is an auxiliary load-balancing loss that pushes routing toward uniformity, but it pulls against the language-model objective and so trades quality for balance. Much of the L2 story (§4) is the slow removal of that tension, culminating in bias-based, auxiliary-loss-free schemes [Wang et al., 2024]. Reading the literature through these three levers shows L1 largely solved, L2 where the modelling gains now sit, and L3 where serving cost—and therefore effort—concentrates. The dimensions we compare in the cross-cutting matrix (Table 3) are the observable projections of these three levers onto each system.

The decomposition is not perfectly clean: granularity and serving interact (finer experts raise all-to-all cost), and a router change can shift the balancing problem. We use the three levers as the primary analytic axis, not as a claim that they are fully independent.

With the levers separated, the rest of the survey takes them in turn and asks, for each, whether it is where the field’s measurable gains now come from—beginning with routing.

3 Routing Strategies (L1)

After an early burst of competing assignment rules—token-choice, expert-choice, hashing, and balanced assignment—routing converged on learned top- k token-choice, and the router is now a settled foundation rather than the locus of the field’s recent gains.

The strongest case against this claim is that the routing alternatives were never decisively beaten on their own terms. Expert-choice routing guarantees perfect balance by construction [Zhou et al., 2022]; balanced assignment frames routing as an optimal-transport problem [Lewis et al., 2021]; even fixed hashing is a competitive baseline [Roller et al., 2021]. If one of these were revived at frontier scale and beat top- k token-choice, routing would again be where the gains hide and our thesis would be wrong.

3.1 Token-choice top- k and the first sparse models

The foundational move routes each token to its top-scoring experts. Lepikhin et al. [2021] establish top-2 token-choice with an auxiliary balancing loss as the workhorse for scaling sequence-to-sequence models, and Fedus et al. [2021] simplify it to top-1 (“Switch”), trading a little quality for halved routing cost and showing the recipe scales to a trillion parameters. Du et al. [2021] carry top-2 token-choice into a decoder-only LM and demonstrate the central economic claim—matching a dense baseline at a fraction of the activated FLOPs. This lineage fixes the template every later model inherits: a learned linear router, a small top- k , and a balancing penalty.

3.2 Expert-choice: inverting the assignment

Zhou et al. [2022] invert the direction of choice: instead of each token picking experts, each expert picks the tokens it wants, up to a fixed capacity. This guarantees perfect load balance and lets “important” tokens receive more compute, but it breaks causality at training time (an expert’s choice depends on the whole batch) and complicates autoregressive decoding. Expert-choice remains influential as an idea, yet frontier autoregressive models did not adopt it—evidence that the balance it buys is now obtainable more cheaply within token-choice.

3.3 Balanced assignment and hashing

Lewis et al. [2021] (BASE) cast routing as a balanced linear-assignment problem solved per batch, removing the need for an auxiliary loss altogether; Roller et al. [2021] go further and route by a *fixed* hash, showing that even a non-learned router is a serious baseline. Both results are double-edged: they prove that balance need not cost quality, which is exactly the lesson that auxiliary-loss-free balancing later reclaims *inside* token-choice, but their assignment machinery did not survive into frontier recipes.

3.4 Stability as the residual routing problem

The one routing issue that did not vanish is training stability. Zoph et al. [2022] diagnose router logits drifting to large magnitudes and add a router z -loss that regularizes them, making sparse models stable and transferable enough to fine-tune. More recent work continues to attack stability geometrically—Cheng et al. [2025] reparameterise the router in an eigenbasis to keep routing stable—confirming that what is left to improve in routing is narrow and special-cased rather than fundamental. Table 1 summarises the family: the assignment rules differ, but every frontier-adopted entry is a small top- k token-choice router whose remaining difficulty is balance and stability, not the choice rule itself.

Routing is not unimportant: a bad router collapses, and the choice of k trades quality against compute on every token. The alternatives also remain live sources of ideas—expert-choice’s per-token compute budgeting, in particular, may yet return. Our claim is not that routing is solved for all time but that it has stopped being the *binding* constraint at frontier scale.

Once the router is a settled top- k token-choice layer, differentiation must come from how the expert pool is structured and balanced—the lever we turn to next.

Table 1: Comparison of Routing Strategies across surveyed works.

Reference	top k	base demo	mechanism	key limitation
Fedus et al. [2021]	1	T5	token-choice top-1	capacity drop / instability
Lepikhin et al. [2021]	2	Transformer MT	token-choice top-2	all-to-all cost
Du et al. [2021]	2	decoder LM	token-choice top-2	training instability
Lewis et al. [2021]	1	Transformer LM	balanced assignment (BASE)	assignment overhead
Roller et al. [2021]	1	Transformer LM	hash routing (fixed)	no learned specialization
Zhou et al. [2022]	variable	decoder LM	expert-choice routing	violates causality at train
Zoph et al. [2022]	2	encoder-decoder	token-choice + z-loss	transfer/fine-tune gaps

4 Granularity, Sharing, and Load Balancing (L2)

The measurable recent wins in MoE come from fine-grained and shared-expert granularity together with auxiliary-loss-free balancing—not from adding more experts—so at a fixed active budget, *how* the expert pool is structured matters more than *how many* experts it contains.

The strongest case against this claim is that frontier models keep raising expert counts as they improve: a model with 256 routed experts beats one with eight, so raw count plainly still helps. If quality tracked total expert count at a fixed active budget—rather than tracking granularity and balancing—then “add experts” would remain the highest-leverage move and our thesis would be wrong.

4.1 Fine-grained experts and shared experts

Dai et al. [2024] make the decisive structural change: split each expert into several smaller ones (fine-graining) so the router can compose specialised combinations, and reserve a few always-on *shared* experts to absorb common knowledge that would otherwise be duplicated across the pool. At a fixed active-parameter budget this combination beats a small number of large experts, because it raises the number of *routing combinations* without raising per-token compute—so “more experts” helps only through granularity, not through raw count. The distinction matters for reading model cards: “more experts” helps when it means finer granularity at the same active budget, and barely helps when it merely inflates the total count.

4.2 The scaling law of granularity

Krajewski et al. [2024] put the intuition on a curve, deriving scaling laws in which granularity is an explicit variable and showing an optimal granularity that rises with compute—so the field’s drift toward many small experts is predicted, not incidental. The same curves expose the catch: past an optimum, finer experts raise routing and all-to-all overhead faster than they raise quality, which is why granularity is a tuned quantity rather than a direction to push without limit.

4.3 From auxiliary loss to loss-free balancing

For years, balance was bought with an auxiliary loss that pulls against the language-model objective, trading quality for uniformity. Wang et al. [2024] remove the tension: instead of a loss, they maintain a per-expert bias that is nudged up for under-used experts and down for over-used ones, balancing load without adding a gradient term that degrades quality. Han and Zhong [2026] supply the analysis, characterising when this bias-based scheme converges to balanced routing—turning a

deployment trick into an understood mechanism. This is precisely the recipe-level, measurable win that raw expert count no longer delivers, and it is the modelling change frontier models adopted fastest.

4.4 Reusing dense checkpoints: sparse upcycling

Granularity also changed *how* MoE models are trained. Komatsuzaki et al. [2022] “upcycle” a trained dense checkpoint into an MoE by cloning its feed-forward block into experts, recovering most of the cost of training sparsity from scratch; He et al. [2024] extend upcycling to fine-grained pools. Upcycling matters to our thesis because it decouples the gains of granularity from a from-scratch training budget, making the L2 recipe cheap to adopt—another sign that the modelling lever is commoditizing. The cross-cutting matrix (Table 3) places these methods beside the production models of the next section; read down the balancing column, the field moves from auxiliary loss to loss-free within three years. Table 2 details the family.

Table 2: Comparison of Granularity, Sharing, and Load Balancing across surveyed works.

Reference	granularity	balancing	experts total active	key limitation
Dai et al. [2024]	fine-grained + 2 shared	aux loss	fine-grained + shared	routing overhead
Krajewski et al. [2024]	fine-grained (laws)	aux loss	granularity G swept	overhead at high G
Wang et al. [2024]	fine-grained	auxiliary-loss-free (bias)	inherits base	new, under-analysed
Han and Zhong [2026]	n.a.	loss-free (analysis)	theory	assumptions
Cheng et al. [2025]	fine-grained	eigen-reparam stable	inherits base	extra params
Komatsuzaki et al. [2022]	coarse	aux loss	upcycled from dense	warm-start bias
He et al. [2024]	fine-grained upcycle	aux loss	upcycled	data for upcycling

Granularity is not free: finer experts raise all-to-all traffic and router cost, and the optimal granularity is budget-dependent rather than “as fine as possible”. Nor is raw scale irrelevant—more total parameters do store more knowledge. Our claim is narrower: at a fixed active budget, the measurable gains track granularity and balancing, not count.

Once granularity and balancing are commoditized recipes, the question that decides what actually ships is no longer how to train a good MoE but how to serve one—the lever we turn to next.

5 Inference Systems and Production Recipes (L3)

The binding constraint on frontier MoE is serving-systems efficiency—the memory and communication cost of having all experts resident while only a few are active—not the modelling recipe, which is why the literature’s centre of gravity has moved to expert parallelism and offloading.

The strongest case against this claim is that serving is mere plumbing atop a capability that modelling created: a better model is worth more than a cheaper one, and systems work merely lowers the cost of a fixed artifact without raising its quality. If serving were a solved, amortizable engineering layer, then effort would stay on modelling and our thesis would be wrong.

5.1 Expert parallelism and all-to-all

Because an MoE’s total parameters exceed any single accelerator’s memory, experts are sharded across devices, and every MoE layer incurs an all-to-all exchange to route tokens to the device holding their expert. This communication, not arithmetic, dominates MoE serving at scale. Zhu et al. [2025] disaggregate expert serving to schedule this traffic efficiently, and Balmau et al. [2025]

Table 3: Cross-cutting comparison of 16 systems across 6 dimensions. Cells marked ? are unfilled at scaffolding time.

System	Routing	Experts (total/active)	Balancing	Precision	Quality signal	Primary failure mode
DeepSeek-MoE [Dai et al., 2024]	token-choice top-k	fine-grained + shared	aux loss	BF16	MMLU at fixed active	routing overhead
FG Scaling Laws [Krajewski et al., 2024]	token-choice	granularity G swept	aux loss	BF16	loss vs G (laws)	overhead at high G
Loss-Free Bal. [Wang et al., 2024]	token-choice top-k	inherits base	auxiliary-loss-free (bias)	BF16	quality at fixed bal.	new, under-analysed
Loss-Free (theory) [Han and Zhong, 2026]	bias update	theory	loss-free (analysis)	n.a.	convergence (theory)	assumptions
ERMoE [Cheng et al., 2025]	eigen-reparam route	inherits base	eigen-reparam stable	BF16	stability/quality	extra params
Sparse Upcycling [Komatsuzaki et al., 2022]	token-choice top-2	upcycled from dense	aux loss	BF16	quality per FLOP	warm-start bias
LLM Upcycling [He et al., 2024]	token-choice top-k	upcycled	aux loss	BF16	quality per FLOP	data for upcycling
Switch [Fedus et al., 2021]	token-choice top-1	few large	aux loss	BF16	quality per FLOP	capacity drop / instability
GShard [Lepikhin et al., 2021]	token-choice top-2	few large	aux loss	BF16	BLEU (MT)	all-to-all cost
GLaM [Du et al., 2021]	token-choice top-2	64 experts	aux loss	BF16	few-shot NLU	training instability
BASE Layers [Lewis et al., 2021]	balanced assignment (BASE)	large	balanced by design	BF16	LM perplexity	assignment overhead
Hash Layers [Roller et al., 2021]	hash routing (fixed)	large	fixed (hash)	BF16	LM perplexity	no learned specialization
Expert Choice [Zhou et al., 2022]	expert-choice routing	variable cap	balanced by design	BF16	quality per FLOP	violates causality at train
ST-MoE [Zoph et al., 2022]	token-choice + z-loss	large	aux + z-loss	BF16	transfer quality	transfer/fine-tune gaps
Mixtral [Jiang et al., 2024]	token-choice top-2	8x7B / 13B act	aux loss	BF16	MMLU 70.6	memory of 47B total
DeepSeek-V3 [DeepSeek-AI et al., 2024b]	token-choice top-k	671B / 37B act	auxiliary-loss-free	FP8	MMLU 88.5	serving memory

attack the same bottleneck through expert sharding—both are systems papers about moving tokens, not about modelling.

5.2 Offloading and memory-constrained inference

The standing-memory cost is sharpest off the datacenter. Eliseev and Mazur [2023] keep only active experts on the GPU and stream the rest from CPU memory, with a cache that exploits the skew in expert use. Xue et al. [2024] make MoE servable on personal machines via activation-aware prefetching. Kamahori et al. [2024] orchestrate CPU and GPU jointly, and Cao et al. [2024] pipeline computation and transfer for high throughput under tight memory. The sheer volume and recency of this work—offloading, prefetching, sharding, mixed-precision expert caches—is itself the evidence: the field is spending its effort where the binding constraint is.

5.3 Production recipes: reverse-engineering the frontier

Frontier production models reveal this convergence most directly of all. Jiang et al. [2024] ship an open 8×7B model that activates roughly 13B parameters per token, demonstrating that a commoditized top-2 token-choice MoE is competitive with much larger dense models. DeepSeek-AI et al. [2024a] pair fine-grained MoE with a compressed attention cache to serve a 236B-total / 21B-active model economically, and DeepSeek-AI et al. [2024b] stack the full L2 recipe—fine-grained plus shared experts, auxiliary-loss-free balancing, and FP8 training—into a 671B-total / 37B-active model. Muennighoff et al. [2024] reproduce the recipe openly (1B-active / 7B-total), and Shen et al. [2024] show it can be trained cheaply. Read across these systems, the modelling recipe is the *same*; what differs, and what each report dwells on, is how to train and serve it within a memory and communication budget.

5.4 Convergence: from modelling to serving

The pattern is unmistakable in Figure 2: the early milestones are modelling milestones (Switch, GShard, GLaM), the middle ones are recipe milestones (DeepSeek-MoE, loss-free balancing), and the most recent cluster is almost entirely serving systems—the literature’s centre of gravity has moved from designing experts to moving them. A field whose frontier models share a recipe and whose new papers are about moving experts is a field whose binding constraint has migrated from modelling to systems. Table 4 separates the recipe layer from the serving layer across these works.

Table 4: Comparison of Inference Systems and Production Recipes across surveyed works.

Reference	layer	model change	optimizes	key idea
Jiang et al. [2024]	recipe	8x7B top-2	open MoE quality	sparse MoE at 13B active
DeepSeek-AI et al. [2024b]	recipe	fine-grained+shared, FP8	frontier train+serve	loss-free balancing, 37B ac...
DeepSeek-AI et al. [2024a]	recipe	MLA + fine-grained MoE	economical MoE	21B active / 236B total
Muennighoff et al. [2024]	recipe	fine-grained, 64 experts	open reproducible MoE	1B active / 7B total, open
Shen et al. [2024]	recipe	sparse attn+FFN MoE	cheap MoE training	Llama2 quality at \$0.1M
Qwen et al. [2024]	recipe	fine-grained + shared	production MoE	stacked routing+balance+serve
Eliseev and Mazur [2023]	serving	none	consumer-GPU inference	expert offloading + cache
Yuan et al. [2024a]	serving	none	offload inference	efficient expert offloading
Xue et al. [2024]	serving	none	personal-machine MoE	activation-aware prefetch
Zhu et al. [2025]	serving	none	large-scale MoE serving	disaggregated expert serve
Cao et al. [2024]	serving	none	memory-constrained MoE	high-throughput pipeline
Kamahori et al. [2024]	serving	none	CPU-GPU MoE inference	CPU-GPU expert orchestration

Systems work does not raise a model’s quality; it lowers the cost of serving a fixed model, and in that narrow sense the steelman is right. New modelling ideas can still reset the frontier. Our claim is that, given a commoditized modelling recipe, serving efficiency is what decides which MoE models can be deployed at all.

If serving is the binding constraint and the modelling recipe is shared, then honest progress must be measured at a fixed active and memory budget—which raises the evaluation question we take up next.

6 Evaluation: Quality per Active FLOP (L2/L3)

The honest axis for comparing MoE models is quality per *active* FLOP at a fixed budget—not raw expert count or total-parameter count—and benchmarks that do not control for active parameters systematically over-credit raw size.

The strongest case against this claim is that total-parameter scaling laws still predict quality: a model that stores more knowledge answers more questions, regardless of how few parameters it activates per token, so total count is a fair axis after all. If quality tracked total parameters as cleanly as it tracks active FLOPs, then reporting headline total size would be defensible and our metric would be needless pedantry.

6.1 Why active FLOPs, not total parameters

The entire value proposition of MoE is that total and active parameters *diverge*: the same active budget can back wildly different total counts, so a comparison that fixes total count compares models that cost different amounts to run, while a comparison that fixes active FLOPs compares models that cost the same. Du et al. [2021] make the point empirically by matching a dense baseline at a fraction of the activated FLOPs; the comparison is only meaningful because it holds the active budget fixed. Reading the cross-cutting matrix (Table 3) down the “experts (total/active)” column shows why a single headline number misleads: Mixtral’s 47B total and DeepSeek-V3’s 671B total activate 13B and 37B respectively, so their deployment costs are far closer than their total counts suggest.

6.2 Fine-grained scaling laws as the measurement frame

Krajewski et al. [2024] give the field the right instrument: scaling laws with granularity as an explicit variable, which let one ask whether a reported gain comes from more compute, finer granularity, or merely more stored parameters. Without such a frame, a model that wins by activating more FLOPs is indistinguishable from one that wins by being better designed—and the literature routinely conflates the two.

6.3 MoE-specific evaluation pitfalls

Three MoE-specific evaluation pitfalls recur across the literature. First, *count inflation*: reporting total parameters as the model’s “size” credits memory that the token never touches. Second, *contamination*, shared with dense LLMs but sharper here because MoE models are trained on very large corpora where benchmark leakage is likely; Yuan et al. [2024b] stress measuring inference cost under a roofline model precisely so that throughput claims are comparable. Third, *serving-blind quality*: a model that scores well but cannot be served within a memory budget is not, in practice, better—which is why Liu et al. [2024] survey inference optimization as a first-class concern rather than an afterthought. An evaluation that ignores the active-versus-total gap and the serving budget measures the wrong thing.

Active-FLOP accounting is not complete: it ignores the standing memory cost that total parameters impose, which is real and, as §5 argued, often binding. The right report is therefore two numbers—quality per active FLOP *and* total memory footprint—not a single headline. Our claim is only that active FLOPs, not total count, is the axis on which quality should be read.

Once quality is read per active FLOP and against a memory budget, the gains attributable to raw count shrink and those attributable to granularity, balancing, and serving stand out—exactly the reframing the open problems below build on.

7 Challenges and Open Problems

7.1 Serving memory dominates deployment

An MoE’s total parameters must reside in memory even though only a fraction are active, so memory—not active compute—gates what can be deployed. Offloading and prefetching keep only active experts resident [Eliseev and Mazur, 2023, Xue et al., 2024], and disaggregated serving schedules expert traffic [Zhu et al., 2025]. These lower the cost of a fixed model but do not change the standing footprint; total parameters still set the floor on deployable hardware. Co-design granularity with offloading and parallelism so total memory stops gating deployment (FD1).

7.2 Routing instability and collapse

Learned routers drift, collapse onto a few experts, or under-utilise the pool, especially under distribution shift. A router z -loss regularizes logit magnitudes [Zoph et al., 2022], and eigen-reparameterised routing improves stability [Cheng et al., 2025]. These dampen symptoms; the underlying tendency of a softmax router to concentrate mass remains, and robustness under shift is untested. Routers with provable balance and stability guarantees under shift (FD2).

7.3 The balancing-versus-quality tension

Auxiliary load-balancing losses trade language-model quality for uniform load, and the loss-free alternative is new and under-analysed. Bias-based auxiliary-loss-free balancing removes the loss term [Wang et al., 2024], with partial theory for when it converges [Han and Zhong, 2026]. The conditions under which bias-based balancing matches or beats an auxiliary loss are only partly understood, so adoption is empirical. A full theory of loss-free balancing and when to prefer it (FD3).

7.4 Granularity overhead erodes its own gains

Finer experts raise routing and all-to-all cost faster than quality past an optimum, so granularity has a budget-dependent ceiling. Scaling laws locate the optimal granularity [Krajewski et al., 2024], and serving systems cut all-to-all cost [Balmau et al., 2025]. The optimum shifts with hardware and budget, and current kernels make fine granularity expensive enough to cap it below the modelling optimum. All-to-all and grouped-GEMM kernels that make fine granularity nearly free (FD4).

7.5 Evaluation credits raw count

Benchmarks rarely control for active parameters, so a model that simply activates or stores more is over-credited relative to a better-designed one. Roofline-based inference accounting [Yuan et al., 2024b] and inference-optimization surveys [Liu et al., 2024] push toward cost-aware comparison. Headline reports still lead with total parameters, and few results fix the active budget when comparing quality. Benchmarks and reporting norms that fix the active-FLOP budget and report memory alongside quality (FD5).

7.6 Expert specialization is opaque

What individual experts specialise in, and whether that specialization transfers, is poorly understood. Fine-grained and shared-expert designs assume a division of labour [Dai et al., 2024], and upcycling inherits dense structure into experts [Komatsuzaki et al., 2022]. The assumed specialization is rarely measured, so pruning, routing, and sharing decisions rest on folklore rather than evidence. Methods to measure and steer what experts learn (FD6).

8 Future Directions

The open problems above pair one-to-one with the directions below; each direction names a concrete next step rather than a wish.

FD1 — Memory-aware MoE co-design. Building on offloading and disaggregated serving [Eliseev and Mazur, 2023, Zhu et al., 2025], we expect granularity and sharing to be chosen *jointly* with the serving plan, so that the total-parameter footprint is treated as a first-class design constraint rather than a consequence—turning “how big a model fits” from a deploy problem into a training-time objective.

FD2 — Robust, analyzable routers. Because routers still drift and collapse [Zoph et al., 2022, Cheng et al., 2025], the trajectory points toward routers with provable balance and stability under distribution shift, replacing the empirical patches that dominate today.

FD3 — A theory of loss-free balancing. The partial analysis of bias-based balancing [Han and Zhong, 2026] suggests a full account is within reach: characterising exactly when loss-free balancing matches or beats an auxiliary loss would convert the current empirical recipe [Wang et al., 2024] into a principled default.

FD4 — Granularity-overhead-aware kernels. If the optimal granularity is capped by all-to-all cost [Krajewski et al., 2024, Balmau et al., 2025], then kernels that make fine-grained routing nearly free would let models reach the modelling optimum the scaling laws predict but the hardware currently forbids.

FD5 — Active-budget-controlled evaluation. We expect reporting to move from total-parameter headlines to quality at a fixed active-FLOP budget reported alongside memory footprint [Yuan et al., 2024b, Liu et al., 2024]; the honest headline becomes “quality at X active FLOPs and Y GB”, not “ N total parameters”.

FD6 — Interpretable expert specialization. Since fine-grained and shared designs assume a division of labour [Dai et al., 2024], methods that measure and steer what experts learn would put pruning, routing, and sharing decisions on evidence rather than folklore.

9 Conclusion

The rise of Mixture-of-Experts created an illusion that bigger expert pools are the story; this survey has tried to dispel it. MoE’s measurable gains no longer come from raw expert count or hand-tuned load-balancing, and the binding constraint on frontier models has migrated from the modelling recipe to serving systems. Read through the three levers—routing, granularity and balancing, and serving—the field’s last five years tell a single story of a moving bottleneck.

Routing became table stakes. After expert-choice, hashing, and balanced assignment were explored, the field converged on learned top- k token-choice, leaving only stability and balance as residual issues. A router is now something one configures, not something one researches.

The modelling wins concentrated in granularity and balancing. Fine-grained experts with a few shared experts, and auxiliary-loss-free balancing, are the changes that moved quality at a fixed active budget, and both are now cheap, commoditized recipes—reproduced openly and even upcycled from dense checkpoints. “More experts” helps only when it means finer granularity, not larger total count.

Serving economics decide what ships. Because all experts must be stored while only a few are active, expert parallelism, all-to-all scheduling, and offloading are what make a frontier MoE deployable, and the recent literature is overwhelmingly about moving experts rather than designing them—visible convergence rather than a single breakthrough.

Evaluation is the field’s compass, not its scoreboard. The gap between total and active parameters is only legible under active-budget-controlled measurement, which is why a single headline number misleads. The most consequential figures an MoE paper can report are quality per active FLOP *and* total memory footprint; until reporting practice makes that switch, total-parameter headlines will keep overstating capability and misdirecting effort toward the lever—raw count—that is already commoditized.

The bottleneck is memory and systems, not modelling. Two models that share the now-standard granularity-and-balancing recipe can differ enormously in whether they can be served at

all, which locates the causal levers in serving systems and in the memory cost of standing parameters, neither of which a new balancing trick can supply.

A call to action. The productive next moves are to co-design granularity with the serving plan, to put loss-free balancing on a theoretical footing, to build kernels that make fine granularity free, and to report quality at a fixed active budget. This thesis would have to be revised if a model were shown to win, at a fixed active-FLOP budget and memory footprint, purely by increasing total expert count with no change to granularity, balancing, or serving—the one experiment that would put raw scale back at the center. Until then, the honest reading is that the next gains in usable MoE will be won in the serving stack and the balancing rule, not in the size of the expert pool.

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